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ORIGINAL ARTICLE

Rural/urban differences in receipt of governmental rental assistance: Relationship to health and disability

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Abstract

Purpose: Housing is essential to health. Governmental rental assistance is one way to increase access to affordable housing, but little is known about how it varies by rural/urban location. This paper seeks to address that gap by examining rural/urban and within-rural differences in receipt of rental assistance, with particular attention differences by health and disability.

Methods: We used data from the 2021 National Health Interview Survey (n = 28,254) to conduct bivariate analyses to identify significant differences in receipt of rental assistance by rural/urban location. We then conducted logistic regression analyses to generate odds ratios of receiving rental assistance, adjusting for self-rated health, disability, sociodemographic characteristics, and the US Census region.

Findings: When limiting the sample to those who rent (20.6% of rural residents and 29.6% of urban residents), rural residents were nearly 5 percentage points more likely to receive rental assistance (13.1% vs 8.2%, P<.001). Rural recipients of rental assistance were more likely to have a disability than urban residents (27.9% vs 20.3%, P<.05) and were more likely to report fair/poor health (41.6% vs 31.4%, P<.05).

Conclusions: Rural residents are less likely to rent their homes, but, among those who rent, they are more likely to receive governmental rental assistance. This may be reflective of the greater need for rental assistance among rural residents, who were in poorer health and of lower socioeconomic status than urban renters. As housing is essential to good health, policy attention must prioritize addressing a persistent and growing need for affordable housing in rural and urban areas alike.

KEYWORDS
disability, housing, policy, structural drivers of health

Housing is essential to health.[1–4] Housing is associated with one's social environment, environmental exposures, and financial stability, all of which can be directly linked to specific health outcomes.[1] Affordability is one important aspect of housing.[4] Yet, affordable housing is out of reach for many people in the United States, which became especially apparent during the COVID-19 pandemic as policies were enacted to prevent evictions.[5,6] In fact, COVID-19 incidence and mortality rates increased in states where eviction moratoriums had expired.[6]

Housing affordability issues long predate the COVID-19 pandemic, however, and still persist 3 years after the pandemic's onset.[7–9] High housing cost burden is associated with poorer health, and may cause individuals to choose between paying for housing and paying for other necessities, such as food and medical care.[10,11] For some lower-income individuals and families, government rental assistance can be helpful in addressing affordability issues.[12,13] Prior research has found a strong link between receiving rental assistance and positive health outcomes.[13,14] Such improvements in health include decreased stress, better mental health, and lower rates of diabetes and obesity.[14]

Rental assistance comes in various forms, including Section 8 and other US Department of Housing and Urban Development vouchers.[15] Rural, suburban, and urban areas alike have access to federal rental assistance programs, including through public housing developments, housing vouchers, and other multifamily programs.[15] In fact, rural areas are home to one-fifth of all public housing units.[16] Development of public housing units and other affordable housing is incentivized at the federal level by the Low Income Housing Tax Credit, which provides tax incentives for building and renovating affordable housing (rather than a direct benefit to low-income renters or home buyers).[15]

In areas addition to the programs listed above, USDA Rural Development also provides rental assistance with an emphasis on rural populations.[15,17] For example, the Section 515 program provides loans to developers of rental housing for older adults, while the Section 514 program provides loans for the development of rental housing for farm workers. Additionally, the Section 521 program provides project-based assistance to owners of USDA-financed housing or Farm Labor Housing projects for low-income tenants who are unable to pay their rent.[15,17] Eligibility for USDA housing programs depends on Congressionally established definitions of rural-based primarily on population size (generally, between <2,500 people to <20,000 people, depending on the program) and community characteristics (eg, "rural in character," "lack of mortgage credit" in low-income communities).[18] Despite the wide variety in housing programs and rental assistance provided by the federal government, the need for such assistance still far exceeds use across all geographic types.[15] Rural residents are in poorer health and have fewer financial resources overall compared to urban residents.[19,20] Therefore, they may have greater needs for rental assistance. However, very little research has looked at how widely rental assistance is used in rural areas and how the demographics of rental assistance recipients differ in rural and urban areas. Even less research has connected rental assistance in rural and urban areas with health and disability measures.[21] This study addresses that gap by identifying sociodemographic and health correlates of receiving rental assistance among rural and urban residents.

METHODS

Data and sample

Data for this study come from the 2021 National Health Interview Survey (NHIS). The NHIS is an annual, nationally representative survey of adults and children in the United States, which has been fielded since the 1960s. We accessed the data through IPUMS Health Surveys.[22] We limited our analysis to sample adults (18+) as only they were asked some demographic questions, including the question on receiving rental assistance, resulting in a final sample size of 28,254.

Measures

Rental assistance was defined by respondents answering "yes" to "Are you/your family paying lower rent because the Federal, State, or local government is paying part of the cost?" This question was asked of any sample adult who reported that they lived in a house or apartment that was being rented. If necessary, interviewers were able to provide this additional guidance to respondents: "Federal, State, or Local government housing programs for persons with low income may take many forms. Government housing assistance could come from monetary assistance to help pay rent, a program called 'Section 8', direct payments to landlords, vouchers, or other types of assistance from a local housing authority. Living in public housing is considered housing assistance from the government."

Rural was defined using the 2013 NCHS Urban-Rural Classification Scheme, with all nonmetropolitan counties defined as "rural" and all metropolitan counties defined as "urban."[23] To assess health, we used a measure of self-rated health (fair/poor vs good/very good/excellent) and disability. Disability was defined using the Washington Group Short Set Composite Disability Indicator, in which respondents are classified as having a disability if they have "a lot of difficultly" or "cannot do [activity] at all" in the following 6 areas: vision, hearing, walking or climbing steps, communicating in usual language, washing or dress, and remembering or concentrating.[24]

We also included other covariates that are associated with differences in housing access and health outcomes, including gender, age, race and ethnicity, sexual orientation, marital status, employment, and educational attainment. These demographic characteristics are all associated with housing due to structural racism, sexism, and historical and current discriminatory housing and economic policies and practices.[4,25] We did not include income directly, because of its overlap with rental assistance eligibility; however, we did include employment status and educational attainment as proxies for socioeconomic status. Finally, we included the US Census Region (Northeast, North Central/Midwest, South, and West) in order to detect differences in housing and demographic patterns by geographic location.

Analysis

We conducted bivariate analyses, comparing rural and urban adults in the full sample and comparing rural and urban adults who rent their homes, using chi-squared tests to detect significant differences. We then conducted logistic regression analyses to generate odds ratios of receiving rental assistance among renters only, first for the full sample of renters and then stratified by rural/urban location in order to see whether the association with covariates differed by rurality.[26] All logistic regression models were adjusted for the full suite of covariates described above. We used survey weights in all analyses to approximate national estimates and to account for the complex survey design.

RESULTS

Table 1 shows sociodemographic and health characteristics for all rural and urban adults and for rural and urban renters. In the full sample, compared to urban adults, rural adults were older; more likely to be non-Hispanic White or American Indian/Alaska Native; less likely to be lesbian, gay, or bisexual; more likely to be married; and less likely to be employed or to have a college degree. Rural adults were also in poorer health and had higher rates of disability. Rural adults were more likely than urban to live in the Northeast and West, and less likely to live in the North Central/Midwest and South regions. Among adults who rent only, many of these same sociodemographic and health differences remained significant between rural and urban adults. However, statistically significant differences were no longer present by sexual orientation. Rural renters were half as likely to have a college degree in their family, compared to urban renters (18.4% vs 37.3%, $P<.001$). Rural renters were also more than twice as likely as urban renters to have a disability (16.3% vs 8.1%, $P<.001$) and were nearly 10 percentage points less likely to report they were in good/very good/excellent health (76.4% vs 85.7%, $P<.001$).

Rural residents were more likely than urban residents to own their homes (76.4% vs 68.6%, $P<.001$; see Table 2). Among the full sample, less than 3% of all adults received any form of rental assistance, with no statistically significant differences between rural and urban residents. However, when limiting the sample to those who rent, rural residents were nearly 5 percentage points more likely to receive rental assistance (13.1% vs 8.2%, $P<.001$).

In adjusted logistic regression models predicting receipt of rental assistance among renters, rural location remained significantly associated with receiving rental assistance (adjusted odds ratio [AOR]: 1.44, $P<.05$; see Table 3). The odds of receiving rental assistance among all renters in the sample increased with age and were higher for females; those identifying as non-Hispanic Black, American Indian/Alaska Native, and other/multiple races; and people who were divorced, separated, widowed, or never married. The odds of receiving rental assistance were lower for those with higher education, who were employed, and who were in better health. The odds of receiving rental assistance were significantly lower in the North Central/Midwest, South, and West, compared with the Northeast region.

In stratified analyses by rural and urban location, many of the covariates remained consistently associated with receiving rental assistance. However, for rural residents, being Hispanic was associated with lower odds of receiving rental assistance (AOR: 0.06, $P<.05$), and being Asian was associated with higher odds of receiving rental assistance (AOR: 6.46, $P<.05$). Among rural residents, being lesbian, gay, or bisexual was also associated with higher odds of receiving rental assistance (AOR: 3.90, $P<.05$).

DISCUSSION

In this study, we find that rural residents are less likely to rent their homes, consistent with prior research showing higher homeownership rates among rural residents.[20,27,28] However, nearly 10% of renters in this study lived in rural areas, and rural residents who rented were more likely than their urban counterparts to receive governmental rental assistance. Rural renters also reported poorer health, higher rates of disability, and lower rates of education and employment than urban renters. Altogether, these findings may be indicative of a greater need for housing assistance in rural areas, as well as indicative of an intersection between housing and health for rural residents.

We also identified diversity in sociodemographic correlates of rental assistance by rural and urban location. In particular, we found that urban renters who are Black or American Indian/Alaska Native were more likely than their White counterparts to receive governmental rental assistance, whereas neither group had elevated odds of rental assistance in rural areas. Yet, previous research has demonstrated the health impacts of structural racism in rural areas, especially for rural Black and Indigenous (ie, American Indian/Alaska Native) residents.[29,30] Rural Black residents, in particular, were overrepresented among rural renters, indicating an inequity in homeownership rates relative to rural White residents; combined with our findings on receipt of rental assistance, there is likely an opportunity for better connecting rural Black, Indigenous, and other residents of color with housing support. Recent research also shows particularly high rates of housing cost burden among Black rural residents, compared with non-Hispanic White rural residents,[31] and much higher rates of substandard housing (eg, incomplete kitchens and plumbing) among rural Indigenous residents compared with all other rural and urban residents (of any racial or ethnic group).[32] These findings on race, rurality, and housing are especially salient when combined with previous research showing poorer health outcomes among Black and Indigenous rural residents, relative to non-Hispanic White rural residents and relative to all urban residents.[29,30] Addressing housing cost and quality issues at the intersection of rurality and racism is an important step in remedying health inequities.[4]

Further, we found that rural Hispanic renters had significantly lower odds of receiving rental assistance compared with their White counterparts. The same was not true for urban Hispanic residents. Hispanic individuals are the fastest-growing demographic in rural areas[33] and were also disproportionately represented among renters in the rural sample of this study. Prior research has shown inequities in access to health care and services for rural Hispanic adults[34–36]; our research adds to this and shows a need for improving access to opportunities for rural Hispanic residents. Governmental rental assistance programs should ensure that they are provided in a culturally responsive and linguistically inclusive manner, recognizing the unique barriers faced by Hispanic renters in rural areas.[36] Health care and social service providers might collaborate to consider how best to strengthen access to social safety net services for low-income Hispanic rural residents.

Our analyses show inequities in who is receiving rental assistance, overall and by rural and urban location. This adds to previous research showing that the need for rental assistance is greater than the current resources to provide it. For example, a 2017 report showed that nearly 1 million rural residents were receiving rental assistance, but more than twice that (2.3 million) experienced severe housing costs

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TABLE 1 Sociodemographic and health characteristics among all adults and renters, by rural and urban location.

	All adults			Renters		
	Rural	Urban	*P*-value[a]	Rural	Urban	*P*-value[b]
Weighted frequencies	13.3%	86.7%		9.8%	90.2%	
Age			<.001			<.01
18-24	9.6%	11.7%		16.2%	17.6%	
25-34	15.2%	18.0%		24.9%	29.2%	
35-44	14.3%	16.8%		16.1%	18.9%	
45-54	15.9%	15.5%		14.9%	13.3%	
55-64	18.7%	16.2%		15.3%	10.5%	
65+	26.4%	21.7%		12.6%	10.6%	
Female	52.4%	51.6%	.402	52.8%	51.2%	.402
Race and ethnicity			<.001			<.001
Non-Hispanic White	81.0%	60.0%		73.6%	44.7%	
Hispanic	6.0%	18.6%		8.2%	25.4%	
Non-Hispanic Black	7.2%	12.4%		12.0%	19.9%	
American Indian/Alaska Native	3.7%	1.0%		2.4%	1.3%	
Asian	1.0%	6.7%		1.4%	6.7%	
Other/multiple races	1.1%	1.3%		2.4%	2.0%	
Sexual orientation			<.001			.149
Lesbian, gay, bisexual	3.4%	5.3%		6.2%	8.0%	
Heterosexual	96.6%	94.7%		93.8%	92.0%	
Marital status			<.001			<.001
Married/cohabitating	63.9%	59.8%		45.5%	45.9%	
Separated/divorced/widowed	18.8%	15.3%		24.6%	16.6%	
Never married	17.3%	24.9%		29.9%	37.5%	
Employed	55.7%	63.2%	<.001	56.1%	67.9%	<.001
Highest educational attainment in family			<.001			<.001
< High school	6.3%	4.4%		9.2%	7.5%	
High school degree	28.7%	18.4%		38.7%	25.8%	
Some college/2-year degree	33.3%	26.6%		33.7%	29.5%	
College or more	31.7%	50.6%		18.4%	37.3%	
Self-rated health			<.001			<.001
Fair/poor	18.8%	12.8%		23.6%	14.3%	
Good/very good/excellent	81.2%	87.2%		76.4%	85.7%	
Has a disability	13.3%	8.1%	<.001	16.3%	8.1%	<.001
Region			<.001			<.01
Northeast	9.5%	18.7%		12.2%	18.1%	
North Central/Midwest	32.3%	19.0%		28.8%	16.8%	
South	41.3%	37.4%		39.3%	35.7%	
West	17.0%	24.9%		19.8%	29.5%	
Unweighted Ns	4,143	24,111		908	7,381	

Note: Data are from 2021 National Health Interview Survey. *P*-values signify statistically significant differences between rural and urban adults in the [a]full sample and [b]renters only.

TABLE 2 Homeownership status and receipt of rental assistance by rural/urban location.

	Rural	Urban	*P*-value
Homeownership			<.001
Own or being bought	76.4%	68.6%	
Rent	20.6%	29.6%	
Other arrangement	3.0%	1.8%	
Receives rental assistance (all)	2.7%	2.4%	.442
Receives rental assistance (renters only)	13.1%	8.2%	<.001

Note: Data are from 2021 National Health Interview Survey. *P*-values signify statistically significant differences between rural and urban adults.

TABLE 3 Adjusted odds of receiving rental assistance among rural and urban renters.

	Full sample AOR	Rural only AOR	Urban only AOR
Rural	1.44*		
Age (Ref: 18-24)			
25-34	2.28***	1.16	2.52***
35-44	3.31***	2.58*	3.35***
45-54	3.18***	2.81	3.24***
55-64	3.49***	2.04	3.82***
65+	3.95***	1.63	4.55***
Female	1.67***	2.48**	1.57***
Race and ethnicity (Ref: non-Hispanic White)			
Hispanic	1.18	0.06*	1.37
Non-Hispanic Black	2.38***	1.47	2.60***
American Indian/Alaska Native	2.46***	1.27	2.87***
Asian	1.28	6.46*	1.26
Other/multiple races	3.76***	1.46	4.57***
Sexual orientation (Ref: heterosexual)			
Lesbian, gay, bisexual	1.36	3.90*	1.13
Marital status (Ref: married/cohabitating)			
Separated/divorced/widowed	2.72***	2.84**	2.79***
Never married	3.03***	2.47**	3.24***
Employed	0.27***	0.29***	0.26***
Highest educational attainment (Ref: <high school)			
High school degree	0.78	0.65	0.83
Some college/2-year degree	0.66**	0.65	0.67*
College or more	0.26***	0.14***	0.29***
Self-rated health (Ref: fair/poor)			
Good/very good/excellent	0.77***	0.63	0.79
Has a disability	1.19	0.85	1.26
Region (Ref: Northeast)			
North Central/Midwest	0.41***	0.41*	0.41***
South	0.37***	0.41*	0.37***
West	0.44***	0.64	0.42***

Note: Data are from 2021 National Health Interview Survey. Logistic regression models limited to adult respondents who rent their homes (n = 8,242). Adjusted odds ratios (AOR) significant at: *$P<.05$; **$P<.01$; ***$P<.001$.

(paying more than half of their income on housing), but did not receive rental assistance.[15] Given the well-documented relationship between stable housing and health, this should be viewed as a serious public health issue. As the economic and health impacts of the COVID-19 pandemic persist, the need for assistance with affordable housing will likely only continue to grow.[9,37]

Policy action is needed to equitably expand rental assistance programs. Additionally, increasing awareness about the connection between housing affordability and health is an important element to impacting policy action.[38] Further, public support for preventing evictions and addressing rental affordability was high during the COVID-19 pandemic,[37] which might be able to be translated into longer-term policy to address housing affordability.

Limitations

As with any study, our results should be considered in light of their limitations. First, our data are cross-sectional, meaning that we cannot determine the direction of effect between housing, health, and rurality. We are also limited by sample sizes, especially for specific rural racial groups who rent their homes. For example, we find an association between being Asian and receiving rental assistance among rural renters, but that finding is based on a relatively small sample of rural Asian renters. More research is needed to better understand within-rural differences in receipt of rental assistance, especially by age. Finally, the NHIS relies on self-report of rental assistance, which may be subject to bias.[39]

CONCLUSIONS

In this study, we find that rural residents are less likely to rent their homes, but, among those who rent, they are more likely to receive governmental rental assistance than their urban counterparts. This may be reflective of the greater need for rental assistance among rural residents, who were found to be in poorer health and of lower socioeconomic status than urban renters in the sample. Furthermore, we find inequities among rural residents in receipt of rental assistance by sociodemographic characteristics, including age, race, ethnicity, sexual orientation, and education. This may demonstrate the need for expanded and equitable rental assistance efforts to ensure affordable and fair housing for all. As housing is essential to good health, policy attention must prioritize addressing a persistent and growing need for affordable housing in rural and urban areas alike.

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CONFLICT OF INTEREST STATEMENT

The authors report no conflict of interest.

DISCLOSURES

The information, conclusions, and opinions expressed in this manuscript are those of the authors, and no endorsement by FORHP, HRSA, NIH, or HHS is intended or should be inferred.

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